

Introduction to Data Science 2026 – Week 2

Exercise 4: Junk Charts

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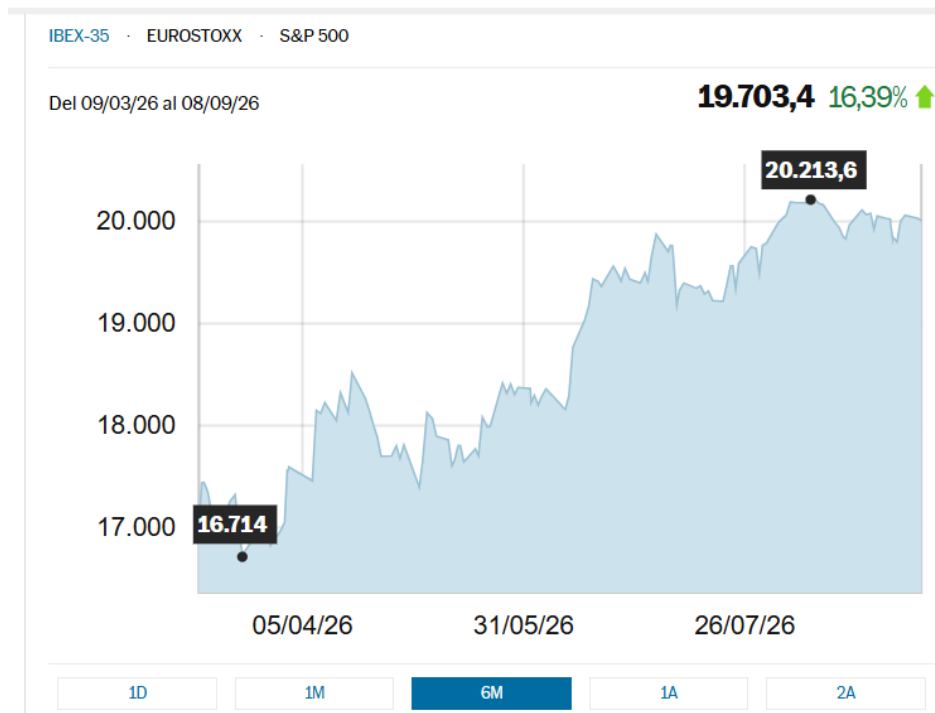
September 9, 2026

This report presents three real-world statistical charts found “in the wild”, each with a different visualization problem, together with an improved version of each and an explanation of what was changed and why.

1 Chart 1: IBEX-35 six-month evolution

Source: stock market data website (screenshot).

1.1 Original

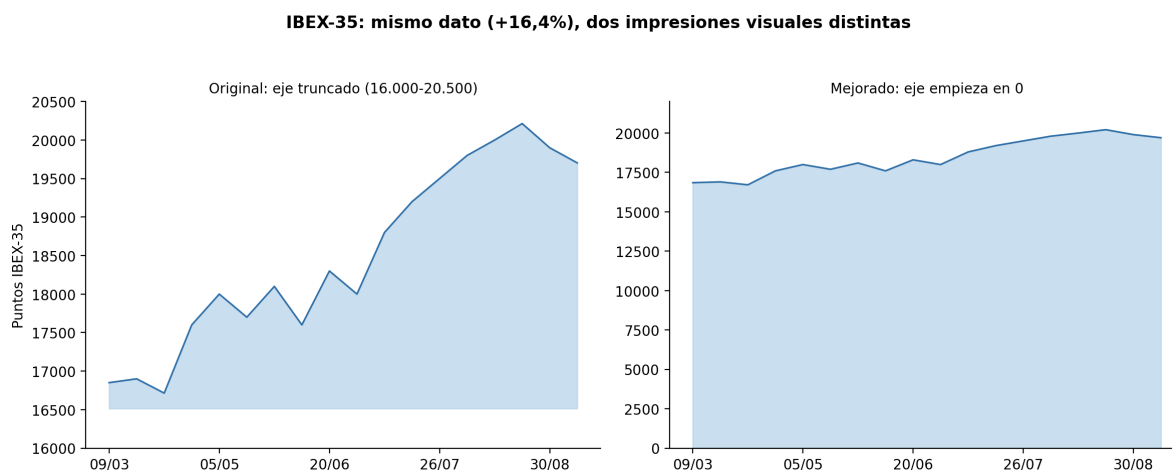


1.2 Problem identified

The y-axis is truncated: it ranges from approximately 16,000 to 20,500 points instead of starting at 0. The index actually rose from 16,714 to 20,213.6 – a real increase of 16.39%, as stated in the chart itself. However, because the visible range of the y-axis is so narrow relative to the values, the line appears to almost double, climbing from near the bottom to near the top of the plot area. A reader who only skims the shape of the curve (without reading the axis labels carefully) is likely to perceive a much larger increase than what actually occurred.

It is worth noting that truncated axes are a common convention in financial charts, since stock indices have no meaningful “zero” and a full 0-based scale would hide the day-to-day volatility that investors care about. Even so, pairing a visually dramatic curve with a large, prominent “+16.39%” label right next to it creates a mismatch between the visual impression and the actual magnitude of the change – this combination is the real issue, more than the truncation alone.

1.3 Improved version



1.4 What was changed and why

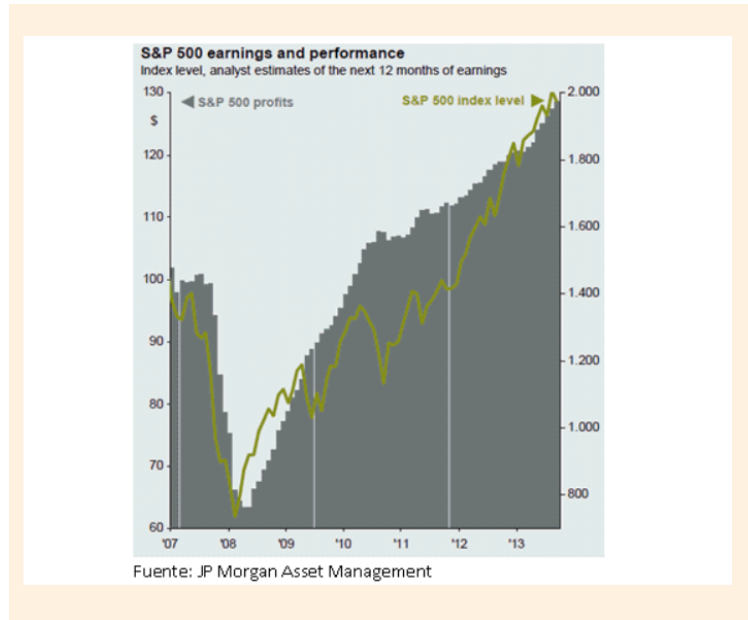
The improved version places the original (truncated-axis) chart side by side with a version whose y-axis starts at 0. This makes the actual magnitude of the increase honestly visible: a real, but far more modest, upward trend, consistent with the +16.39% figure. Showing both versions together also helps the reader understand *why* the two impressions differ, rather than simply hiding the original scale.

Data note: exact daily values were not available, so the underlying series was approximated visually from the original chart at irregular intervals; the overall shape and start/end values closely match the original.

2 Chart 2: S&P 500 earnings and performance (dual axis)

Source: JP Morgan Asset Management chart, reproduced in Expansión’s blog “Confundir lo relativo con lo absoluto” (<https://www.expansion.com/blogs/blog-jsq/2015/01/23/confundir-relativo-con-absoluto.html>).

2.1 Original

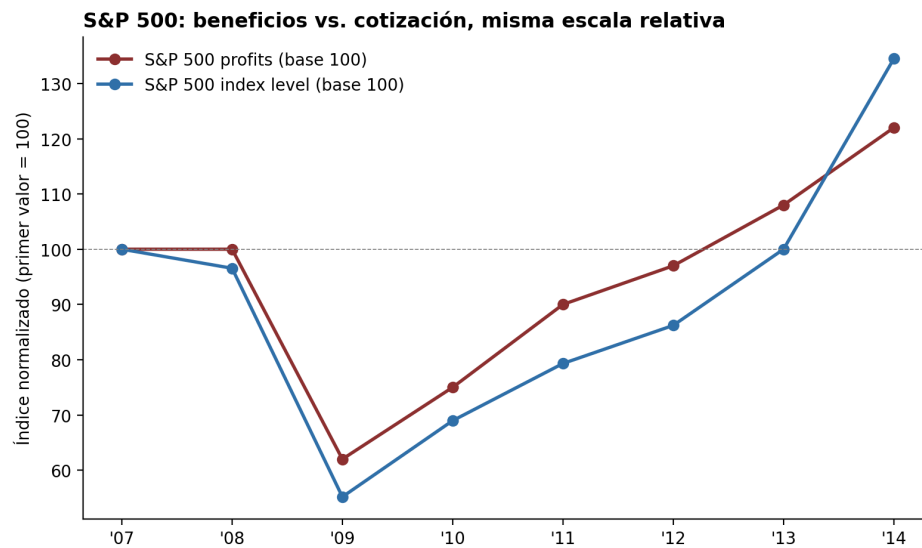


2.2 Problem identified

This chart plots two completely different quantities on two independent y-axes: “S&P 500 profits” (left axis, roughly \$60–130) and “S&P 500 index level” (right axis, roughly 800–2,000 points). Both series are drawn on top of each other and, because each axis range was chosen independently, the two curves appear to move almost in lockstep.

This is a well-known and heavily criticized charting trick: by choosing the range of each axis freely, it is possible to make almost any two time series visually “line up”, regardless of whether they are actually correlated to a similar degree, or whether one is simply tracking the other with a lag or a different growth rate. No correlation coefficient or other statistic is shown – the apparent relationship is purely a visual artifact of the chosen scales, and would look completely different if either axis range were changed.

2.3 Improved version



2.4 What was changed and why

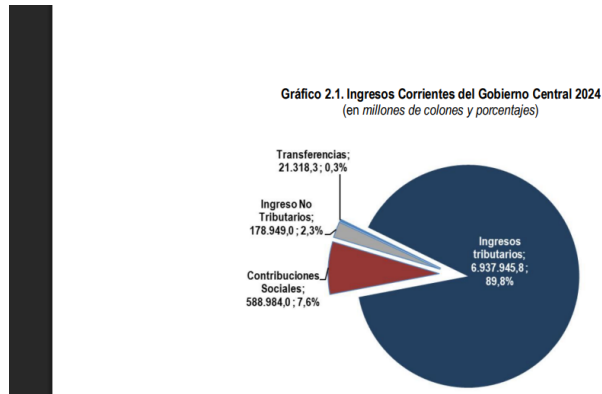
Both series were normalized to a common base of 100 at the first data point (2007) and plotted on a single, shared axis. This removes the arbitrary scaling choice entirely: the two lines can now only appear close together if their *relative* (percentage) changes are actually similar. The normalized chart reveals something the original hid: by 2014, the index level had grown proportionally *more* than underlying profits, suggesting the market may have been pricing in future growth rather than tracking current earnings one-to-one – a nuance that the original dual-axis chart’s “perfectly aligned” curves obscured.

Data note: exact values were not published alongside the chart, so approximate yearly values were read visually from the original image; they reproduce the same overall shape (sharp 2008–09 drop, steady recovery) as the source.

3 Chart 3: Ingresos Corrientes del Gobierno Central 2024 (pie chart)

Source: PDF report, <https://d1qqtien6gys07.cloudfront.net/wp-content/uploads/2023/09/23912.pdf>.

3.1 Original

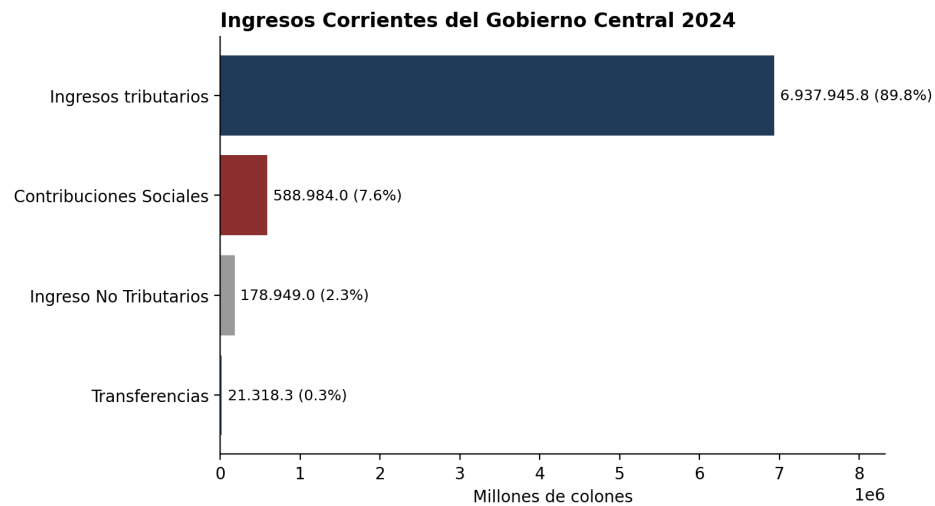


3.2 Problem identified

This pie chart suffers from several compounding issues:

- **Arbitrary “exploded” slice:** the “Contribuciones Sociales” slice (7.6%) is pulled out from the rest of the circle, while the other three slices are not, with no clear reason for singling out that one category. This is a purely decorative choice that adds visual noise without conveying information.
- **Extreme imbalance:** one category (“Ingresos tributarios”) accounts for 89.8% of the total, leaving only 10.2% split across the other three. A pie chart is a poor choice here, since almost the entire circle is a single color and the smaller, arguably more interesting categories are reduced to thin, hard-to-compare slivers.
- **Text doing the chart’s job:** because the small slices are visually illegible, every slice needs an attached text label with the exact value and percentage. If the numbers must be read as text to understand the chart, the visual encoding itself has failed at its main purpose.

3.3 Improved version



3.4 What was changed and why

The same four categories and exact values (read directly from the original chart's labels) are shown as a horizontal bar chart, sorted from largest to smallest. This makes the dominant category still clearly dominant, but now the three smaller categories can actually be compared to each other by eye – something impossible in the original pie chart. Values and percentages are kept as direct labels for precision, but here they *complement* a chart that already communicates the relative magnitudes correctly on its own, rather than substituting for it.